

Qualification Pack



Smart Grid Technician

QP Code: PSS/Q2416

Version: 1.0

NSQF Level: 4

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PSS/Q2416: Smart Grid Technician

Brief Job Description

The incumbent in the job is responsible for operation and maintenance of smart grid. The role of the technician is to undertake both planned and reactive operation and maintenance tasks across the range of smart grid systems.

Personal Attributes

The job holder must have an eye for detail as well as the patience and discipline required to carry out detailed and repetitive tasks. The candidate should be able to read and understand technical manuals, instructions, warnings and adhere to applicable safety rules and regulations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N2409: Carry Out Maintenance Of Smart Grid](#)
2. [PSS/N1720: Perform Operation Of Smart Grid](#)
3. [PSS/N1337: Apply Basic Health and Safety Practices for Power Related Work](#)
4. [PSS/N1336: Working effectively with others](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution
Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113.9900



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Minimum Educational Qualification & Experience	12th grade Pass (With Science Background) OR 10th grade pass (with two years of any combination of NTC/NAC/CITS or equivalent in relevant trade) OR Previous relevant Qualification of NSQF Level (3.5) with 1-2 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant Experience OR Completed 3 year diploma after 10th (in Electrical/Electronics/Instrumentation Engineering OR)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	N/A
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	22/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-04-PW-03317-2024-V1-PSSC
NQR Version	1.0



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PSS/N2409: Carry Out Maintenance Of Smart Grid

Description

This unit is about performing various tasks for routine and preventive maintenance of smart grid.

Scope

The scope covers the following :

- Prepare for maintenance work
- Perform maintenance activities
- Troubleshooting and repairing of faulty equipment
- Perform post-maintenance activities

Elements and Performance Criteria

Prepare for maintenance work

To be competent, the user/individual on the job must be able to:

- PC1.** Determine the routine maintenance requirements of various components and control system by interpreting manufacturer manual, maintenance checklist and schedule, work instructions, etc
- PC2.** Read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance activities
- PC3.** Identify various systems, sub systems and control systems of the smart grid
- PC4.** Arrange tools, consumables, spare parts, etc. required for performing repair and maintenance
- PC5.** Clear the work area thoroughly to ensure no unwanted materials are present before starting the work
- PC6.** Ensure that devices, machines etc. in the smart grid are not connected to any power source before starting maintenance work
- PC7.** Adhere to all safety norms (such as wearing protective gloves and shoes, etc.) in accordance with the international/ national and organisational standards
- PC8.** Inform all affected parties of the intended work plan prior to disconnecting power supply line

Perform maintenance activities

To be competent, the user/individual on the job must be able to:

- PC9.** Perform basic health check-up of instruments, sensors, controllers etc. as specified in the manufacturer manual
- PC10.** Start and shut down process of smart grid to observe the functioning of power plant and its systems under the supervision of supervisor/engineer
- PC11.** Check the technical and mechanical systems to ensure their operation within the specified parameters
- PC12.** Inspect power stations, transmission lines, distribution lines, automated systems and other critical equipment for identification of faults, possible wear and tear etc.



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- PC13.** Locate the protection devices i.e. relay, breaker etc., conduit, cables & other underground system to perform maintenance work
- PC14.** Carry out the maintenance activities by following appropriate techniques and procedures
- PC15.** Monitor running parameter of smart grid and auxiliaries' system as per design standard
- PC16.** Test the system parameters to know abnormal condition of the system and healthiness of connected equipment
- PC17.** Check the problem and keep the superiors informed about progress or any delays in resolving the problem
- PC18.** Upgrade or modify the existing unhealthy equipment/system after confirmation from supervisor/engineer
- PC19.** Suggest technical changes in equipment/system to supervisor/engineer
- PC20.** Suggest software upgradation and testing required to supervisor/engineer

Troubleshooting and repairing of faulty equipment

To be competent, the user/individual on the job must be able to:

- PC21.** Carry out troubleshooting of faulty or unhealthy equipment to determine root causes and develop processes to avoid repeat failure
- PC22.** Identify the type of faults, required tools and devices for removal of faults
- PC23.** Conduct breakdown maintenance and then inspect the equipment for leaks, short circuit in parts, breakage of wires and clamps, vibrations, erratic/problematic performance or any other problem as reported in the complaint
- PC24.** Isolate the faulty part and replace/repair it by following correct procedures and manufacturer specification
- PC25.** Clean and tighten the terminals, screws, clamps and cables and ensure that they are free from dirt, grime and corrosion by following the instructions given in manufacturer manual
- PC26.** Conduct the appropriate tests related to the functional performance of equipment, as required
- PC27.** Coordinate with engineers and process technicians to resolve the issues requiring their assistance
- PC28.** Coordinate with the relevant personnel to evaluate the functioning of smart grid
- PC29.** Assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel

Perform post-maintenance activities

To be competent, the user/individual on the job must be able to:

- PC30.** Leave the work area in clean and tidy condition after the completion of work
- PC31.** Dispose off waste materials such as failed parts/aggregates, as per organisation's and environmental policies
- PC32.** Record all repairs carried out, time taken and other significant findings observed during the work process
- PC33.** Ensure that all maintenance activities are adequately addressed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** Relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** The applicable isolation and lock-off procedures or permit-to-work procedure that applies
- KU3.** How to interpret drawings, specifications, manufacturers manuals and other documents needed during the maintenance process
- KU4.** Process to be adopted to establish the background of the fault
- KU5.** Maintenance activities for a smart grid
- KU6.** How to evaluate various types of information available for fault diagnoses such as operator-provided information, equipment self-diagnosis, and testing instrument measurements
- KU7.** Use of various testing and fault diagnostic tools and equipment
- KU8.** How to analyse evidence and evaluate possible characteristics and causes of specific faults/problems
- KU9.** How to relate to the previous reports/records of similar fault conditions
- KU10.** How to evaluate the likely risk of running the equipment with the displayed fault, and the effects the fault could have on health and safety, and on the overall process or system
- KU11.** The process of handling and maintaining smart grid systems
- KU12.** Relevant precautions to be taken to prevent Electrostatic Discharge (ESD) damage to electronic circuits and components
- KU13.** Working purpose of individual units/components and how they interact
- KU14.** Importance of ensuring the control systems is isolated or put into manual control, and appropriate trip locks
- KU15.** How to carry out visual checks of the smart grid systems including checking for leaks, security of joints and physical damage
- KU16.** Relevant replacement parts, materials and other items required for the maintenance process
- KU17.** Relevant techniques used to dismantle/assemble integrated equipment i.e. release of pressures/force, proof marking to aid reassembly, plugging exposed pipe/component openings, dealing with soldered joints, screwed, clamped and crimped connections
- KU18.** Importance and process of attaching identification marks/labels to removed components or cables, to assist with reassembly
- KU19.** Applicable methods for checking components are fit for purpose, and the need to replace electronic modules, sensors, transmitters, transducers, electronic boards and other failed items
- KU20.** Ratings and specifications of cables, fuses, switches and wires
- KU21.** How to resolve the common problems encountered during the maintenance of the smart grid systems
- KU22.** The importance of making process improvements by engaging the relevant stakeholders
- KU23.** Process of carrying out different types of maintenance such as preventive, corrective, predictive and reactive maintenance, and the activities these entail

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Note the information related to work and processes
- GS2.** Write reports and observations related to work in English/regional language



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- GS3.** Read and interpret and process flowchart for all operations
- GS4.** Read symbols, manuals and operation documents to understand the Equipment used into operation
- GS5.** Discuss task lists, schedules and activities with the seniors and team members
- GS6.** Follow organization rule-based decision making process
- GS7.** Take decisions with systematic course of actions and/or response
- GS8.** Plan and organize tasks to meet deadlines
- GS9.** Find ways of modifying difficult operating stages to make it operation friendly
- GS10.** Apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** Analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** Recognise a workplace problem and take suitable action to resolve it



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for maintenance work</i>	7	9	-	-
PC1. Determine the routine maintenance requirements of various components and control system by interpreting manufacturer manual, maintenance checklist and schedule, work instructions, etc	1	1	-	-
PC2. Read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance activities	1	1	-	-
PC3. Identify various systems, sub systems and control systems of the smart grid	2	2	-	-
PC4. Arrange tools, consumables, spare parts, etc. required for performing repair and maintenance	1	1	-	-
PC5. Clear the work area thoroughly to ensure no unwanted materials are present before starting the work	1	1	-	-
PC6. Ensure that devices, machines etc. in the smart grid are not connected to any power source before starting maintenance work	-	1	-	-
PC7. Adhere to all safety norms (such as wearing protective gloves and shoes, etc.) in accordance with the international/ national and organisational standards	-	1	-	-
PC8. Inform all affected parties of the intended work plan prior to disconnecting power supply line	1	1	-	-
<i>Perform maintenance activities</i>	18	26	-	-
PC9. Perform basic health check-up of instruments, sensors, controllers etc. as specified in the manufacturer manual	2	3	-	-
PC10. Start and shut down process of smart grid to observe the functioning of power plant and its systems under the supervision of supervisor/engineer	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Check the technical and mechanical systems to ensure their operation within the specified parameters	2	3	-	-
PC12. Inspect power stations, transmission lines, distribution lines, automated systems and other critical equipment for identification of faults, possible wear and tear etc.	2	4	-	-
PC13. Locate the protection devices i.e. relay, breaker etc., conduit, cables & other underground system to perform maintenance work	1	1	-	-
PC14. Carry out the maintenance activities by following appropriate techniques and procedures	2	3	-	-
PC15. Monitor running parameter of smart grid and auxiliaries' system as per design standard	1	2	-	-
PC16. Test the system parameters to know abnormal condition of the system and healthiness of connected equipment	2	3	-	-
PC17. Check the problem and keep the superiors informed about progress or any delays in resolving the problem	1	1	-	-
PC18. Upgrade or modify the existing unhealthy equipment/system after confirmation from supervisor/engineer	1	1	-	-
PC19. Suggest technical changes in equipment/system to supervisor/engineer	1	1	-	-
PC20. Suggest software upgradation and testing required to supervisor/engineer	1	1	-	-
Troubleshooting and repairing of faulty equipment	11	20	-	-
PC21. Carry out troubleshooting of faulty or unhealthy equipment to determine root causes and develop processes to avoid repeat failure	2	3	-	-
PC22. Identify the type of faults, required tools and devices for removal of faults	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Conduct breakdown maintenance and then inspect the equipment for leaks, short circuit in parts, breakage of wires and clamps, vibrations, erratic/problematic performance or any other problem as reported in the complaint	2	4	-	-
PC24. Isolate the faulty part and replace/repair it by following correct procedures and manufacturer specification	1	2	-	-
PC25. Clean and tighten the terminals, screws, clamps and cables and ensure that they are free from dirt, grime and corrosion by following the instructions given in manufacturer manual	1	2	-	-
PC26. Conduct the appropriate tests related to the functional performance of equipment, as required	1	2	-	-
PC27. Coordinate with engineers and process technicians to resolve the issues requiring their assistance	1	2	-	-
PC28. Coordinate with the relevant personnel to evaluate the functioning of smart grid	1	2	-	-
PC29. Assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel	1	1	-	-
<i>Perform post-maintenance activities</i>	4	5	-	-
PC30. Leave the work area in clean and tidy condition after the completion of work	1	1	-	-
PC31. Dispose off waste materials such as failed parts/aggregates, as per organisation's and environmental policies	1	1	-	-
PC32. Record all repairs carried out, time taken and other significant findings observed during the work process	1	2	-	-
PC33. Ensure that all maintenance activities are adequately addressed	1	1	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2409
NOS Name	Carry Out Maintenance Of Smart Grid
Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution, Erection, Installation and Commissioning - Distribution Downstream
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



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PSS/N1720: Perform Operation Of Smart Grid

Description

This unit is about performing activities related to operation, monitoring and controlling of smart grid components i.e. power stations, transmission lines, and distribution lines under the guidance of supervisor/engineer and as per the specified standards and procedures.

Scope

The scope covers the following :

- Perform operation of smart grid

Elements and Performance Criteria

Perform operation of smart grid

To be competent, the user/individual on the job must be able to:

- PC1.** Follow safe working practices in accordance with the organizational standards and regulations
- PC2.** Obtain instructions from the supervisor/engineer related to day's work
- PC3.** Use specialized equipment, software, and protocols to ensure the safe and stable operation of power systems, as well as responding to emergencies and disturbances
- PC4.** Check the smart grid components for any faults or abnormalities
- PC5.** Ensure smooth operation of power stations, transmission lines, and distribution lines
- PC6.** Monitor load, grid stability, frequency, load dispatch, feeder loading etc.
- PC7.** Monitor and control the status and parameters of smart grid components, such as voltage, frequency, power flow, and demand.
- PC8.** Operate and switch smart grid devices
- PC9.** Operate and monitor critical system such as SAS, RTU, Remote Monitoring Unit (RMU) and other automation system installed in the grid
- PC10.** Coordinate and communicate with other operators, engineers, distributors, and customers in smart grid operation
- PC11.** Implement and follow operating procedures for smart grid operation
- PC12.** Identify and report system faults, events, and anomalies
- PC13.** Restore and maintain power system service in case of outages or contingencies
- PC14.** Suggest technical changes in equipment/system to supervisor/engineer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Relevant legislation, standards, policies, and procedures followed in the organization



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- KU2.** Importance of using personal protective equipment (PPE) against possible electrical hazards as described in the organizational health and safety guidelines and relevant regulations
- KU3.** Application of basic principles of electricity in energy meters
- KU4.** Electrical units used to measure energy outputs, for example KVA, KWH, etc.
- KU5.** Relevant terms, signs, symbols and other graphical representations and their respective interpretations
- KU6.** CEA regulations for smart grid operations
- KU7.** Common power plant terminology e.g. temperature, pressure, ph, current, voltage, resistance, kilowatt (kw), kilowatt hour, plf
- KU8.** Layout and schematic of smart grid including machine arrangement, equipment layout, switchyard, auxiliary arrangements
- KU9.** Various smart grid components and sections i.e. power stations, transmission lines, distribution lines etc.
- KU10.** Operation of various smart grid components and sections
- KU11.** Benefits of smart grid
- KU12.** Features and function of various automation and IoT systems installed in smart grid
- KU13.** Tools, tackles and safety gadgets required in grid
- KU14.** Load management, system stability, frequency, load dispatch, etc
- KU15.** Emergency restoration procedures
- KU16.** How to read technical manuals, blueprints, schematics, diagrams, plans, specifications
- KU17.** How to recognize and report inaccurate work instructions and documentation to designated personnel

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Note the information related to work and processes
- GS2.** Write reports and observations related to work in English/regional language
- GS3.** Read and interpret and process flowchart for all operations
- GS4.** Read symbols, manuals and operation documents to understand the Equipment used into operation
- GS5.** Discuss task lists, schedules and activities with the seniors and team members
- GS6.** Follow organization rule-based decision-making process
- GS7.** Take decisions with systematic course of actions and/or response
- GS8.** Plan and organize tasks to meet deadlines
- GS9.** Find ways of modifying difficult operating stages to make it operation friendly
- GS10.** Apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** Analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** Recognise a workplace problem and take suitable action to resolve it



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform operation of smart grid</i>	30	70	-	-
PC1. Follow safe working practices in accordance with the organizational standards and regulations	1	2	-	-
PC2. Obtain instructions from the supervisor/engineer related to day's work	1	2	-	-
PC3. Use specialized equipment, software, and protocols to ensure the safe and stable operation of power systems, as well as responding to emergencies and disturbances	4	7	-	-
PC4. Check the smart grid components for any faults or abnormalities	2	5	-	-
PC5. Ensure smooth operation of power stations, transmission lines, and distribution lines	2	5	-	-
PC6. Monitor load, grid stability, frequency, load dispatch, feeder loading etc.	2	5	-	-
PC7. Monitor and control the status and parameters of smart grid components, such as voltage, frequency, power flow, and demand.	3	7	-	-
PC8. Operate and switch smart grid devices	3	7	-	-
PC9. Operate and monitor critical system such as SAS, RTU, Remote Monitoring Unit (RMU) and other automation system installed in the grid	3	8	-	-
PC10. Coordinate and communicate with other operators, engineers, distributors, and customers in smart grid operation	2	5	-	-
PC11. Implement and follow operating procedures for smart grid operation	2	5	-	-
PC12. Identify and report system faults, events, and anomalies	2	5	-	-
PC13. Restore and maintain power system service in case of outages or contingencies	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Suggest technical changes in equipment/system to supervisor/engineer	2	5	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1720
NOS Name	Perform Operation Of Smart Grid
Sector	Power
Sub-Sector	
Occupation	Operation and Maintenance - Generation
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



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PSS/N1337: Apply Basic Health and Safety Practices for Power Related Work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly

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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1337
NOS Name	Apply Basic Health and Safety Practices for Power Related Work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N2409.Carry Out Maintenance Of Smart Grid	40	60	-	-	100	35
PSS/N1720.Perform Operation Of Smart Grid	30	70	-	-	100	35
PSS/N1337.Apply Basic Health and Safety Practices for Power Related Work	39	61	-	-	100	10
PSS/N1336.Working effectively with others	34	56	-	-	90	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	163	277	-	-	440	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.